

Why Are Organisations Moving to Microservices?

Microservices is a software architecture style in which complex applications are composed of small, independent, narrowly-focused processes communicating with each other using language independent APIs. Amazon and LinkedIn have both implemented this architecture, and others are following suit.

The word 'Microservices' emerged from a workshop on software architecture in May 2011, when, for the first time, architecture style was discussed. Very simply put, Microservices is like UNIX commands, which are designed to do one single thing but can be piped together to perform some complex operations. We can say that Microservices is loosely coupled, fine-grained, follows the Single Responsibility Principle [Robert C. Martin], is highly cohesive and autonomous. A more specific and clear definition by Martin Flower is, "The Microservices architectural style is an approach to developing a single application as a suite of small services, each running in its own process and communicating with lightweight mechanisms, often an HTTP resource API. These services are built around business capabilities and are independently deployable by fully automated

deployment machinery. There is a bare minimum of centralised management for these services, which may be written in different programming languages and use different data storage technologies."

Service oriented architecture (SOA), continuous integration and deployment, domain driven development and automation have all led to the Microservices architecture. Netflix, Amazon, LinkedIn, Groupon, PayPal, Airbnb, The Guardian and many other companies have implemented or are moving towards the Microservices architecture.

To understand Microservices better, we need to understand the macro or monolithic application style first. According to Wikipedia, "A monolithic application describes a single-tiered software application in which the user interface and data access code are combined into a single program from a single platform." A monolithic application is built or packaged as a single

